

RESEARCH ARTICLE

Dynamic State Estimation of Power Systems Using a Modified Unscented Kalman Filter and Wavelet Transform Considering Missing and Noisy PMU Data

SUBIN KOSHY¹, V. SREEHARI VENUGOPAL², (Member, IEEE),
R. SUNITHA¹, (Senior Member, IEEE), ELIZABETH P. CHERIYAN¹, (Senior Member, IEEE),
AND K. ANANTHAKRISHNAN¹, (Student Member, IEEE)

¹Department of Electrical Engineering, National Institute of Technology Calicut, Kozhikode, Kerala 673601, India

²Delta Electronics India Pvt., Ltd., Bangalore, Karnataka 560048, India

Corresponding author: R. Sunitha (rsunitha@nitc.ac.in)

ABSTRACT The power system is a quasi-static system which experiences change in its state continuously. The advent of Phasor Measurement Units (PMUs) has brought another dimension to wide area monitoring, protection and control (WAMPAC) through the process of dynamic state estimation. However, the existence of noise or missing PMU data due to failure of the measuring unit or communication channel can result in the mal-operation of the control system and also result in erroneous state estimation. This paper proposes a simplified yet robust solution for estimating the dynamic state of the power system under missing PMU data or noisy conditions. In order to obtain PMU data which reflects the real time scenario of the power system, the PMU is modelled in MATLAB Simulink in accordance with IEEE C37.118 standard. The proposed algorithm is based on de-noising technique using the Wavelet Transform (WT) and dynamic state estimation (DSE) technique based on a Modified Unscented Kalman Filter (Modified UKF). This approach improves the robustness of the classical UKF based DSE algorithm, in the absence of PMU data without introducing extra computational complexity. The proposed method, when combined with wavelet-based de-noising, demonstrated superior performance by significantly reducing the estimation errors for both angular speed (ω) rotor angle (δ), thereby clearly outperforming the conventional UKF method. Other simulation results on the standard IEEE 14-bus test system also validate the superiority of the proposed algorithm under various missing and noisy PMU conditions.

INDEX TERMS Dynamic state estimation, phasor measurement unit, synchrophasor data, unscented Kalman filtering, wavelet transform.

I. INTRODUCTION

Globally, power systems are confronting a period of critical change driven mainly by the increased integration of renewable energy sources. The need for quality and uninterrupted power supply has stressed the necessity of employing Wide Area Monitoring Systems (WAMS). Before the introduction of PMU technology, Supervisory Control and Data Acquisition (SCADA) systems were utilized for the monitoring of the power system. However, their data reporting rates

were inappropriate for dynamic state estimation [1]. PMU's faster data reporting rates, together with active dynamic state estimation techniques has enabled real time monitoring and control of the power system. The IEEE standard C37.118.1-2011 describes PMU as a functional unit that can measure positive sequence voltage and current phasors, frequency, and rate of change of frequency (ROCOF) for all possible working circumstances [2]. Time synchronized PMU data is obtained by time stamping the measured data using Global Positioning System (GPS) clock as reference. This synchronized data can be used for real-time operations as well as offline planning applications to improve the power

The associate editor coordinating the review of this manuscript and approving it for publication was Ugur Guvenc¹.

system's operational reliability [3]. Sufficient samples of PMU data are required for the research and designing of wide-area monitoring tools for power systems. Obtaining PMU data from utilities with various power system events for testing and validating proposed method is cumbersome, because of the privacy and security norms set by utility authorities that own and operate the PMUs. Hence, a PMU is modelled in MATLAB Simulink for the real-time generation of sufficient PMU data. It provides a hazard-free environment to simulate a suitable power system model under various test conditions. The core of the PMU Model is the DFT Module, which is used for phasor computation [4], [5].

Monitoring the state of a system on a dynamic basis is the pre-requisite for real-time monitoring of any system. Dynamic State Estimation (DSE) is a technique that is used to approximate the unknown time variant system state variables. The need for real-time monitoring of dynamic state variables of the modern power system is driven by increasing system complexity and the growing demand for reliable and stable grid operations [6]. The power system's dynamic state vector includes generator rotor angles and generator speeds. DSE provides a better picture of the power system dynamics that helps in efficient monitoring of the power system and initiates appropriate protection and control strategies, which would help in improved oscillation monitoring and advanced dynamic security assessment schemes [7]. This helps in efficient monitoring of the power system and initiates appropriate protection and control strategies, which would help in improved oscillation monitoring and advanced dynamic security assessment schemes. The Weighted Least Squares (WLS) technique initially employed for dynamic state estimation [8], [9], [10], is prone to convergence issues and reduced accuracy in ill-conditioned systems, limiting its application for real-time dynamic state estimation tasks [11]. To address these limitations, Kalman filter-based methods were introduced, which use a two-step cycle of prediction and correction to estimate system states [12]. This led to the development of variants such as the Extended Kalman Filter (EKF) [13], Ensemble Kalman Filter (EnKF) [14] and Unscented Kalman Filter (UKF) [15] to handle the non-linearity and stochastic nature of power system measurements. In [16], the authors used the EKF technique, while in [17], a UKF-based method was used for real-time state estimation of a single-machine infinite bus system. Similarly, EnKF [18], and an extended Particle Filter [19] have been explored for DSE, however, both these filters require significantly more computational resources compared to UKF, particularly in case of large power systems. While both EKF and UKF offer better computational efficiency than other filters, UKF generally outperforms EKF in nonlinear scenarios, as identified in [20]. However, classical UKF methods fail significantly in the presence of missing PMU data. To address the aforementioned challenges, the proposed work presents a modified version of the UKF algorithm to improve the robustness of the classical UKF technique during missing PMU data. EKF-UI (Extended Kalman Filter with

Unknown Inputs) is employed in [21] on a reduced third order model of a DFIG (Doubly Fed Induction Generator), which efficiently estimates the dynamic state utilizing phasor measurements from the stator and rotor terminals. In [22] a robust UKF was designed to handle the effect of noise in measurements, but the practical implications of missing data was not included.

While [23] addresses missing data using EKF modelled on a fourth-order generator model, it lacks generalization and does not offer a comprehensive solution to real-time DSE under such constraints. The works in [21], [22], and [23] are validated on standard test systems as well. The data which is recorded by the PMU is bound to have noise associated with it [24]. In [25] and [26], the authors modelled noise as Gaussian distribution having mean as zero and a standard deviation of 0.002 and 0.02 respectively. The standard deviation for the noise in voltage and current data in [27] is considered as 0.005 – 0.01 pu. Higher signal to noise ratio (SNR) results in better estimation. In [28], an SNR of 92 dB is considered optimal to match the field PMU data with simulated PMU data. Nevertheless, no empirical attempt was made to check the distribution of noise and its parameters.

PMUs are usually installed in strategic monitoring points such as high voltage (HV) transmission lines, load dispatch centers, generating stations, etc. The increase in new energy integration and the associated inverter based resources has necessitated the need for real time monitoring and control of the system that resulted in the deployment of PMUs in medium and low voltage (LV) grids. This makes it even more important to remove the noise content in PMU data at various levels of voltage. An efficient de-noising algorithm must be used for the retrieval of original data from the noisy signal. The proposed work uses Wavelet Transform (WT) based de-noising techniques to effectively eliminate noise from the contaminated PMU data without affecting the original signal. Using de-noised data for state estimation results in higher accuracy than using raw noisy inputs.

This paper aims to present a simple yet effective approach to enhance the robustness of the UKF-based Dynamic State Estimation (DSE) algorithm. Since the classical UKF algorithm does not take the scenario of absence of measurement data in to consideration, a Modified UKF algorithm is proposed in this paper. The proposed method focuses on minimizing the error associated with dynamic state estimation caused by the absence of PMU data, without incurring additional computational cost. The Wavelet Transform (WT) de-noising techniques is employed in the proposed approach to effectively filter out the noise from the contaminated PMU data, ensuring minimal distortion to the original signal. The main contributions of the proposed method, by addressing the limitations of the existing literature, are as follows.

- Existing literature often rely on limited quantity of real PMU data for training purposes. In contrast, the proposed work models a PMU in MATLAB Simulink according to IEEE C37.118 standards, enhancing the practicality and relevance of the simulation results.

- Traditional filtering techniques are not optimized to handle measurement noise in PMU data effectively, often leading to inaccurate estimates. The proposed approach incorporates a Wavelet Transform-based denoising algorithm that significantly removes the noise content.
- Many existing state estimation techniques either become computationally complex or fail to function accurately during multiple missing PMU data; however, the proposed UKF based method retains computational efficiency as well as estimation accuracy under cases of both single and multiple missing PMU measurements.

The rest of the paper is structured as follows. Section II reviews the PMU technology and discusses the challenges faced. Section III introduces the proposed state estimation system that incorporates the effects of various erroneous conditions. Section IV provides the results pertaining to the application of the proposed module on a test system. Section V discusses the performance of the proposed scheme, and the paper concludes with Section VI.

II. PMU TECHNOLOGY AND CHALLENGES FACED

A phasor measurement contains phasor data – magnitude (RMS) and phase angle. The differences in angle and voltage magnitude are used for calculating the real and reactive power flows respectively. When the phasor data is time-stamped with the universal satellite clock, it is termed a Synchrophasor. The time stamping not only allows the synchronization of the PMU data which helps in monitoring the power system on a wide area basis but also enables the comparison of data from PMUs at various distant locations. A PMU has three functional parts – measurement, phasor computation, and communication modules. The measurement module consists of anti-aliasing filters, which are used for attenuating frequencies above the Nyquist frequency. The computation module comprises ADCs, GPS receiver and the processor, which computes the phasor. The communication module streams the computed data in the IEEE C37.118 format. The phasor extraction using standard DFT may pose leakage effects which can add error in the phasor estimates. Hence post processing of DFT phasor is to be done to mitigate the drift effects to improve the accuracy [29], [30] to meet the performance requirements as per C37.118.1a2014 standard [31]. PMU needs to send the measured data through an electrical or optical communication channel. Standard data reporting rates (in frames per second) for a PMU operating in a system frequency of 50 Hz are 10, 25 or 50 frames per second, and for a system frequency of 60 Hz, it can be 10, 12, 15, 20, 30 or 60 frames per second. The electrical communication network is preferred for data transfer within a substation, and for a geographically far location, an optical communication network is preferred. The two main standards for PMU data transmission are C37.118 and IEC 61850-90-5 [32]. The measured data from the PMU is transmitted to Phasor Data

Concentrator (PDC) for further data processing. The PMU communicates the data to PDC's and processing units through the communication channel. The PDC gathers phasor data from different PMUs or PDCs and communicates it to other applications [33], mainly signal processing applications. Thus, the failure of a PMU or the communication channel results in the discontinuation of PMU data transfer and can lead to severe issues if not properly identified. The monitoring and state estimation algorithms can process missing data as a fault and can generate incorrect control strategies that need to be prevented. The dynamic state estimation algorithms mentioned in the literature were mainly based on EKF [34], [35], EnKF [14] and UKF [15] techniques. These algorithms produced an accurate estimate of the state variables when the system model was well known, and all the PMU measurements were available. However, it may not hold for a practical power system, which can be affected by communication failures or cyber-attacks, which disrupts the working of the PMU. This is the main motivation for the work, to make the dynamic state estimation of the power system to be more robust.

The measured data can get polluted by noise from the measurement side as well as from the communication side [36]. The data recorded by the PMU is used for monitoring, protection and control purposes, such as event diagnosis, oscillation monitoring, state estimation, etc. When it is communicated to the PDC, and later for data processing, the communication noise also gets added to it. Adding noise to the signal changes the phasor parameters depending on the magnitude of noise. It can result in the mal-operation of the protection or control equipments, which may cause serious damage to the power system as well. Currently, there is no specific IEEE standard which states the characteristic parameters or range of the noise associated with the PMU data. However, in [37], the noise has been assumed to be random having zero mean and a standard deviation of 0.02pu or 30 dB, and in [24], it was approximated as 45 dB for all PMU data. The proposed DSE module effectively eliminates the effect of noise and estimates the state accurately, even during missing PMU Data.

Obtaining real time PMU data from utilities with various power system events for testing and validating the proposed method is cumbersome because control, and monitoring centers often do not disclose the measurement data due to privacy and security concerns. Hence, in this work the PMU modelled in MATLAB Simulink which can provide sufficient PMU data at the desired sampling frequency and under various simulated test conditions. The obtained PMU data is processed using the proposed processing algorithms to estimate the state. The proposed Dynamic State Estimation Module is explained in the following section.

III. PROPOSED DSE MODULE

Fig 1 shows the schematic of the proposed DSE module. The PMU data for the proposed module is collected using a set of PMUs implemented on the standard IEEE 14-Bus Test

system. The simulated PMU model in MATLAB Simulink for the proposed work is intended for the dynamic state estimation purpose which requires speedy response and minimal delay times [30]. In the PMU model, the DFT based phasor estimation technique is used, and it may be due to the narrow range of frequency variation following the disturbances is considered, significant change in estimation errors were not observed with Smart DFT (SDFT). Therefore, the DFT based phasor estimation technique used in this work is found to be in compliance with the C37.118.1a2014 Standard which includes the modified performance requirements of C37.118.1-2011 standard [2]. The model adds Gaussian white noise of 45 dB into the measurement channels to replicate real world PMU device and measurement noise. PMU measurement latency is assumed to be zero in this study, as modeling communication and processing delays falls beyond the scope of the current work. Simulation step size is selected as 0.02 s, which matches the reporting interval commonly adopted in control centers for PMU data. Phasor data is time-stamped with reference to the system time, thereby simulating the time-synchronized PMU measurements. Time stamping is done on the phasor data aligned with system time, simulating time-synchronized PMU data. The estimated phasor data is then sent via the communication channel to the processing unit. The processing unit consists of the proposed algorithms – the WT based de-noising and Modified UKF based state estimation algorithms. The WT based de-noising algorithm de-noises the measured PMU data and effectively estimates the state of the system even under the existence of missing PMU data. The descriptions for the different units in the proposed DSE module are provided in the following subsections.

A. PMU MODEL

The Simulated PMU model collects the analog data from the connected bus of IEEE 14-Bus system. The PMU unit steps down the analog input, routes it through the analog filter unit, and then digitizes it with an analog to digital converter (ADC). The sample and hold module samples the signal and locks the signal value for a particular period. The DFT module computes the corresponding phasor for the analog current and voltage signals. The frequency measurement module evaluates the frequency of the input signal. The derivative module estimates the rate of change of frequency (ROCOF). The PMU data includes namely current and voltage phasors, frequency, and ROCOF, are then transferred to a data system for further analysis [33]. The PMU uses the DFT technique to obtain the phasors from the discretized data. The Discrete Fourier Transform of the signal over a sampling time interval ΔT is given in (1) – (3). The Fourier coefficient X_k is given as

$$X_k = \frac{1}{\sqrt{2}} \frac{2}{N} \sum_{n=0}^{N-1} x(n\Delta T) e^{-j\frac{2\pi kn}{N}}. \quad (1)$$

where N represents the number of sampling points. Taking its trigonometric equivalent,

$$X_k = \frac{1}{\sqrt{2}} \frac{2}{N} \sum_{n=0}^{N-1} x(n\Delta T) \left(\cos\left(\frac{2\pi kn}{N}\right) - j \sin\left(\frac{2\pi kn}{N}\right) \right) \quad (2)$$

Simplification of this expression leads to two terms, namely, the real part X_{kc} and the imaginary part X_{ks} .

$$X_{kc} = \frac{1}{\sqrt{2}} \frac{2}{N} \sum_{n=0}^{N-1} x(n\Delta T) \cos\left(\frac{2\pi kn}{N}\right) \quad (3)$$

$$X_{ks} = \frac{1}{\sqrt{2}} \frac{2}{N} \sum_{n=0}^{N-1} x(n\Delta T) \sin\left(\frac{2\pi kn}{N}\right) \quad (4)$$

Converting (3) and (4) to the polar form gives the magnitude (RMS) and the angle of the phasor quantity under measurement. In order to test the performance of the developed PMU model, the measuring units are placed in the generator bus of standard IEEE-14 bus system for obtaining the measurement data. However, the optimal PMU placement is not considered in this work as it is beyond the scope of the study. The single line diagram of the IEEE-14 bus system used for testing the model is shown in Fig.2.

The measurements obtained from the PMU located at Bus-1 is shown in Fig 3. This figure shows the three phase voltage waveform at Bus-1, phasor voltage magnitude and angle measured by PMU at phase A of Bus-1. For test purposes, a voltage sag of 0.2 pu is initiated between 1s to 1.5s, and a voltage swell of 0.2 pu is initiated between 1.5s to 2s. This PMU Model can also be interfaced with an RTDS simulation testbed, which can aid in reliable PMU data extraction from a larger system on a real time basis.

B. WAVELET TRANSFORM (WT) BASED DE-NOISING

In Fig 3, the recorded PMU data does not contain noise since it is simulated data. However, in the practical scenario, noise gets added along with the PMU data, which may arise from the measurement side as well as the communication side, which can affect the state estimation. The noise can be removed with the help of data processing techniques, such as Fourier transform or WT techniques. However, the Fourier Transform technique has certain disadvantages when processing non – stationary signals. The local characteristic of the wavelet-based de-noising method makes it an efficient technology to remove noise from non – stationary signals [38]. The algorithm developed by Donoho [39] is adopted for the wavelet-based de-noising technique. The mother wavelet was selected from the Daubechies wavelet family (db4). The main steps involved in de-noising include the wavelet transformation of the noisy signal, soft thresholding and then obtaining the inverse wavelet transform to obtain the de-noised signal. The threshold is set and the wavelet coefficients above and below the threshold is

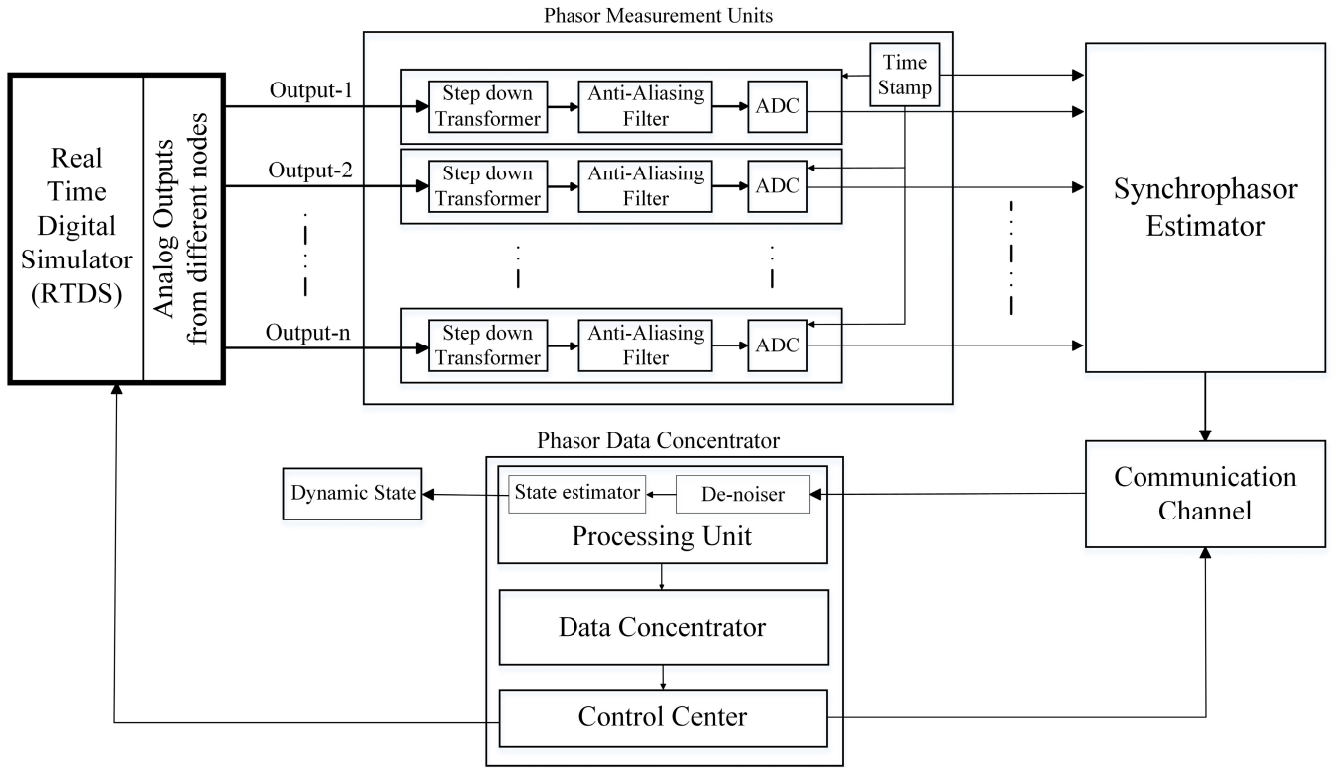


FIGURE 1. Block diagram of the proposed DSE module.

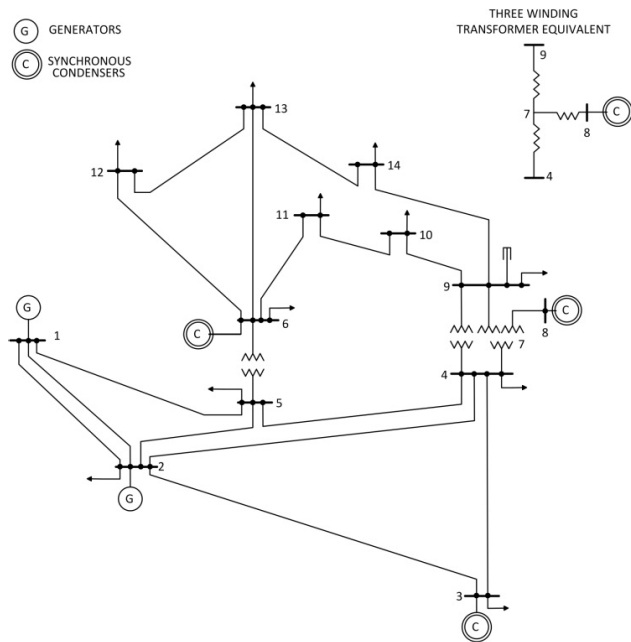


FIGURE 2. IEEE-14 bus system [46].

adjusted and reconstructed according to (5) [39].

$$\tilde{w}_{j,i} = \begin{cases} \text{sgn}(w_{j,i}) (|w_{j,i}| - \lambda_j) & : \text{if } |w_{j,i}| \geq \lambda_j \\ 0 & : \text{if } |w_{j,i}| < \lambda_j \end{cases} \quad (5)$$

where $w_{j,i}$ stands for the wavelet coefficients, $\tilde{w}_{j,i}$ stands for the wavelet coefficients after thresholding and λ_j is the threshold.

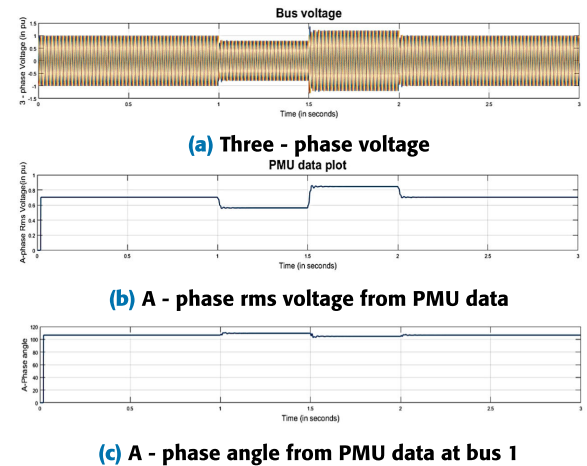


FIGURE 3. Voltage signal obtained from PMU located at Bus-1.

In this work, the measured PMU data is polluted with random Gaussian noise with a magnitude 45 dB [24], and the noisy signal is fed on to the wavelet-based de-noising algorithm which eliminates the noise content of the signal to produce the denoised signal. The denoised signal is used for state estimation purposes and is found to produce a better

estimate compared to the noisy signal. The waveform in Fig 4a shows the signal with noise, and Fig 4b shows the de-noised signal. The SNR ratio of the de-noised waveform is calculated around 40dB, which indicates that the algorithm is able to overcome the effect of noise and produce a de-noised signal.

This de-noised data has been used for state estimation purposes and is observed to yield more accurate estimate of state variables compared to the noisy data. The Modified UKF based DSE algorithm along with system modelling is presented in the following section.

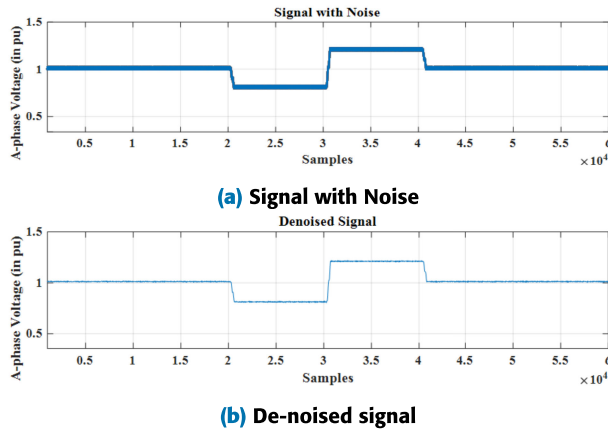


FIGURE 4. PMU data after WT based De-noising technique.

C. MODIFIED UKF BASED STATE ESTIMATION

The static state of an electric power system is a vector that comprises the magnitudes and phase angles of bus voltages. This concept was introduced in the early 1970s by Schweppe [40], [41], [42]. Since then, researchers all over the world have proposed many methods for state estimation. The SCADA systems fail to track the dynamic changes due to their relatively low sampling rate, and hence they are limited to static state estimation only. The higher sampling rate and accuracy of the PMU data help estimate the power system's dynamic state. DSE based algorithms are important, especially during disturbance monitoring, as it ensures the first step towards the power system's reliable operation. The case of missing PMU data is another situation that needs to be taken care of. The DSE models use the present and previous state to estimate the next instant's state. The drawback of the classical UKF based DSE technique is that, during the absence of PMU data, the algorithm detects the same as a fault, which will result in the undesirable operation of the protection and control equipment. The discontinuity in PMU data leads to inaccurate state estimation outputs that may resemble those observed during actual system faults. This can lead to activation of protection schemes such as distance relays, overcurrent protection, or generator protection, which depend on real-time DSE outputs as they may falsely interpret these state estimates as genuine power system faults. This can lead to unintended tripping of breakers, load shedding,

or isolation of healthy equipment. These responses degrade system reliability and may initiate unnecessary emergency control actions.

The proposed work makes use of a Modified UKF based DSE technique for optimally estimating the generator state variables δ (generator rotor angle) and ω (generator rotor speed). Simulation studies have been carried out in IEEE 14 bus test system to demonstrate the robustness and effectiveness of the proposed work.

In this work, time domain simulation is used to generate PMU data samples in compliance with the C37.118.1a2014 Standard. The PMU reporting rate is considered as 20 samples per second. A fourth order transient model also known as two-axis model of the synchronous generator is used for time domain simulation of IEEE 14 bus system [43], [44]. The differential and algebraic equations associated with the two-axis generator model with IEEE-type exciter is presented in (6)-(8).

$$\begin{aligned}
 \frac{d\delta}{dt} &= \omega - \omega_0 \\
 \frac{d\omega}{dt} &= \frac{1}{2H}(P_m - P_e - D(\omega - \omega_0)) \\
 \frac{dE'_q}{dt} &= \frac{1}{T'_{d0}}(E_{fd} - E'_q - (X_d - X'_d)I_d) \\
 \frac{dE'_d}{dt} &= \frac{1}{T'_{q0}}(-E'_d + (X_q - X'_q)I_q) \\
 V_d &= V_T \sin(\delta - \theta_T) \quad V_q = V_T \cos(\delta - \theta_T) \\
 I_d &= \frac{E'_q - V_q}{X'_d} \quad I_q = \frac{V_d - E'_d}{X'_q} \\
 P_e &= V_d I_d + V_q I_q \quad Q_e = -V_d I_q + V_q I_d
 \end{aligned} \tag{6}$$

where P_m is the mechanical power input, P_e is the electrical power output, H is the inertia constant in s, ω_0 is the rated angular velocity in rad/s, δ is the generator rotor angle in radian and D is the damping coefficient in p.u. E'_q , E'_d are q-axis and d-axis transient internal voltages of the generator, V_q , V_d , I_q and I_d are q-axis and d-axis stator voltages and currents respectively. T'_{d0} and T'_{q0} are the open circuit d-axis and q-axis transient time constants in s. P_e and Q_e are the active and the reactive power injected by the generator. V_T and θ_T are generator terminal bus voltage magnitude and phase angle. E_{fd} is the generator internal field voltage. X_q , X_d , X'_q , and X'_d are the q-axis and d-axis reactance under steady state and transient conditions. At steady state, for a lossless system, P_m is assumed to be constant and equal to the real power injection P_e .

Dynamic equations associated with the IEEE-Type I exciter are given follows [43]:

$$\begin{aligned}
 T_e \frac{dE_{fd}}{dt} &= V_R - (K_e + S_E(E_{fd}))E_{fd} \\
 T_f \frac{dR_f}{dt} &= -R_f + \frac{K_f}{T_f} E_{fd} \\
 T_A \frac{dV_R}{dt} &= -V_R + K_A(V_{ref} - V_T - R_f)
 \end{aligned} \tag{8}$$

where V_{ref} is the exciter reference voltage, V_R is the scaled output of the amplifier, K_e and T_e are the self-exciter gain and time constant, K_A and T_A are the exciter amplifier gain and time constant, K_f and T_f are the exciter feedback gain and time constant, R_f is the field resistance.

The loads are modeled as constant shunt admittance y_L as given in (9) where S_{L_i} and V_{L_i} are the complex power and voltage at i^{th} load bus.

$$y_{L_i} = \frac{S_{L_i}^*}{|V_{L_i}|^2} \quad (9)$$

The i^{th} generator's internal voltage and angle ($|E_i| \angle \delta_i$) [45] are calculated using (10).

$$|E_i| \angle \delta_i = \bar{V}_{T_i} + j \frac{X_{d_i}(P_{e_i} - jQ_{e_i})}{\bar{V}_{T_i}^*} \quad (10)$$

The network model for multi machine power system for transient analysis is represented by the relationship between generator current injection vectors and the internal voltage [45], given as,

$$I_n = (Y_{nn} - Y_{ns}Y_{ss}^{-1}Y_{sn})E_n = \hat{Y}E_n \quad (11)$$

where $\hat{Y}$ represents the reduced admittance matrix, I_n is the current injected at the internal node of the generator and E_n is the internal voltage of the generator, n is the number of synchronous generators. Y_{nn} , Y_{ns} , Y_{ss} and Y_{sn} are the elements of bus admittance matrix for transient stability analysis.

The classical UKF system equations corresponding to the state vectors of (6) and (7) are represented in (12) and (13) as,

$$x_{k+1} = f(x_k, k) + w_k \quad (12)$$

$$z_k = h(x_k, k) + v_k \quad (13)$$

where x_k represents the discrete state vector at instant k , the discrete measurement vector z_k includes the current and voltage phasors obtained from the PMU measurements at instant k , $w_k \sim N(0, Q_k)$ is the Gaussian process noise at instant k with covariance Q_k , and $v_k \sim N(0, R_k)$ represents the Gaussian measurement noise at instant k having a covariance R_k . The value of Q , R and error covariance matrix are also assumed with the standard deviation of $\sigma = 10^{-2}$.

The UKF-DSE algorithm is based on the time update and measurement update equations [46]. The state and error covariance matrices are initialized as in (14) and (15).

$$\hat{x}_0^+ = \mathbf{E}(x_0) \quad (14)$$

where $\hat{x}_0^+$ represents the initial state vector.

$$P_0^+ = \mathbf{E}[(x_0 - \hat{x}_0^+)(x_0 - \hat{x}_0^+)^T] \quad (15)$$

where P_0^+ represents the initial error covariant matrix. The time update formulas are represented in (16) – (21) based on the $2n$ 'sigma points'. Here n represents the number of states. The equations are as follows.

1) The $2n$ sigma points $\hat{x}_{(k-1)}^{(i)}$ can be calculated using the following equations,

$$\hat{x}_{k-1}^{(i)} = \hat{x}_{k-1}^+ + x_{*}^{(i)}, \quad i = 1, 2, \dots, 2n \quad (16)$$

where

$$x_{*}^{(i)} = \left(\sqrt{nP_{k-1}^+} \right)_i^T, \quad i = 1, 2, \dots, n \quad (17)$$

$$x_{*}^{(n+i)} = - \left(\sqrt{nP_{k-1}^+} \right)_i^T, \quad i = 1, 2, \dots, n \quad (18)$$

Here $\hat{x}_{k-1}^+$ and P_{k-1}^+ represent the current best approximation of mean and covariance of x_k .

2) The above sigma points are transformed using the nonlinear system equation,

$$\hat{x}_k^{(i)} = f(\hat{x}_{k-1}^{(i)}, k-1) \quad (19)$$

The a priori state estimate $\hat{x}_k^-$ at time k using $\hat{x}_k^{(i)}$ vectors are given by

$$\hat{x}_k^- = \frac{1}{2n} \sum_{i=1}^{2n} \hat{x}_k^{(i)} \quad (20)$$

The a priori error covariance matrix P_k^- is given by

$$P_k^- = \frac{1}{2n} \sum_{i=1}^{2n} (\hat{x}_k^{(i)} - \hat{x}_k^-)(\hat{x}_k^{(i)} - \hat{x}_k^-)^T + Q_{k-1} \quad (21)$$

where Q_{k-1} represents the process noise. Similarly, the measurement update equations are represented in (22) – (26) on the basis of the $2n$ sigma points

$$\hat{x}_k^{(i)} = \hat{x}_k^- + x_{*}^{(i)}, \quad i = 1, 2, \dots, 2n \quad (22)$$

where

$$x_{*}^{(i)} = \left(\sqrt{nP_k^-} \right)_i^T, \quad i = 1, 2, \dots, n \quad (23)$$

and

$$x_{*}^{(n+i)} = - \left(\sqrt{nP_k^-} \right)_i^T, \quad i = 1, 2, \dots, n \quad (24)$$

Here $\hat{x}_k^-$ and P_k^- are the current best approximation for the mean and covariance of x_k . Transforming the sigma points using the nonlinear system equation

$$\hat{z}_k^{(i)} = h(\hat{x}_k^{(i)}, k) \quad (25)$$

The predicted measurement vector is given by

$$\hat{z}_k = \frac{1}{2n} \sum_{i=1}^{2n} \hat{z}_k^{(i)} \quad (26)$$

PMU data can go missing in cases of malfunctioning of PMU, cyber-attacks, or communication failures. It is clear from the classical UKF equations that the DSE algorithm uses the predicted measurements to optimally estimate the state vector. During the absence of PMU data, the error between the actual measurements received and predicted measurements

will be very high and will result in error in the final estimation of system states. The inaccurate state estimation can result in incorrect control decision by the system operators.

This problem is addressed in the proposed work by modifying the classical UKF algorithm. In the proposed approach, the predicted measurement vector $\hat{z}_k^i$ replaces the corresponding measurement data when the data from any PMU is found to be missing. If the measurement data from j^{th} PMU is missing, the replaced measurement vector can be represented as

$$(z_k)_j = (\hat{z}_k)_j \quad (27)$$

By proceeding in this manner, the elements of the error covariance matrix P_z given by (28) and the cross-covariance matrix P_{xz} given by (29) are least effected due to the absence of data. Eventually the Kalman Gain matrix represented as K_k , which is derived from P_z and P_{xz} as per (30) will generate the accurate state estimate.

The error covariance matrix related to the system model and measurements is given by

$$P_z = \frac{1}{2n} \sum_{i=1}^{2n} (\hat{z}_k^{(i)} - \hat{z}_k) (\hat{z}_k^{(i)} - \hat{z}_k)^T + R_k \quad (28)$$

where the measurement noise is represented by R_k . The cross-covariance matrix is given by

$$P_{xz} = \frac{1}{2n} \sum_{i=1}^{2n} (\hat{x}_k^{(i)} - \hat{x}_k) (\hat{z}_k^{(i)} - \hat{z}_k)^T \quad (29)$$

The Kalman Gain matrix is calculated as

$$K_k = P_{xz} P_z^{-1} \quad (30)$$

The estimated error covariance matrix is calculated as

$$P_k^+ = P_k^- - K_k P_z K_k^T \quad (31)$$

The estimate of the updated state is given as

$$\hat{x}_k^+ = \hat{x}_k^- + K_k (z_k - \hat{z}_k) \quad (32)$$

For validating the robustness of the proposed DSE algorithm, simulation studies were conducted on IEEE 14 bus test system for various scenarios as presented in the following section.

IV. SIMULATION RESULTS AND DISCUSSIONS

IEEE 14-Bus test system consists of 5 generators, 17 transmission lines and 3 transformers. System data is taken from [46]. Table 1 specifies the dynamic data of generators used in this study. The system is considered to have a base of 100 MVA and an operating frequency of 60 Hz.

During simulations, different scenarios such as bus fault and line outage were considered for testing the Modified UKF technique's estimation efficiency. Dynamic state variables such as the generator rotor angle δ and generator rotor speed ω are estimated using the proposed approach. PMUs measure the generator terminal voltage and current phasors and

TABLE 1. Generator data for IEEE-14 bus system.

Parameter	BUS 1	BUS 2	BUS 3	BUS 6	BUS 8
X_d (pu)	1.97	1.97	1.97	1.97	1.97
X_q (pu)	1.89	1.89	1.89	1.89	1.89
X'_d (pu)	0.3027	0.3027	0.3027	0.3027	0.3027
X'_q (pu)	0.49	0.49	0.49	0.49	0.49
D (pu)	0.2	0.2	0.2	0.2	0.2
H (s)	7.6	7.6	7.6	7.6	7.6
T'_{d0} (s)	6.84	6.84	6.84	6.84	6.84
T'_{q0} (s)	1	1	1	1	1

calculate the associated real and reactive power injections, using [7]. With the help of the previous state estimate, the incoming measurement, and the power system model, the state estimation process predicts the state vector for the next instant of time. The PMU data is purposefully polluted by adding measurement and communication noise to it. As per IEEE C37.118.1-2011 standard, the noise is considered as Gaussian with zero mean and standard deviation of 10^{-2} . The PMU data is then fed into a de-noising algorithm which removes the noise content in it. The time step for simulation is chosen as 0.02 s, taking into consideration the PMU reporting rate of 50 frames per second (fps) according to IEEE C37.118. It is noted that for waveform-level modeling and internal phasor estimation processes, a finer time resolution would be required. The transient state is modelled by removing and restoring the transmission line connected between bus-4 and bus-5 at $t = 2$ s and $t = 2.2$ s, respectively. In this work, three cases are considered for state estimation. The first case estimates the state when all PMU measurements are present. Case 2 and Case 3 are carried out with one PMU measurement missing and two PMU measurements missing, respectively. The actual state is obtained from the time domain simulation using Runge Kutta numerical integration technique for the system model presented in (6) and (8).

A. CASE 1-ALL PMU MEASUREMENTS AVAILABLE

In this case, estimation using the classical UKF DSE technique is performed. Since there is no missing PMU data, the classical UKF DSE technique's performance is not affected. Hence, the proposed method and the classical UKF DSE technique yield the same estimate in this case. Fig 5a estimation of rotor speed ω , whereas Fig 5b shows the estimation of absolute rotor angle δ at bus 1 of the IEEE-14 bus system.

B. CASE 2-ONE OF THE PMU MEASUREMENT IS MISSING

In this case, the data from the PMU connected at bus 3 is purposefully cut off from $t = 6$ s. Practically, this situation is likely to occur in a real power system due to the failure of a PMU or failure of communication channel, or due to the

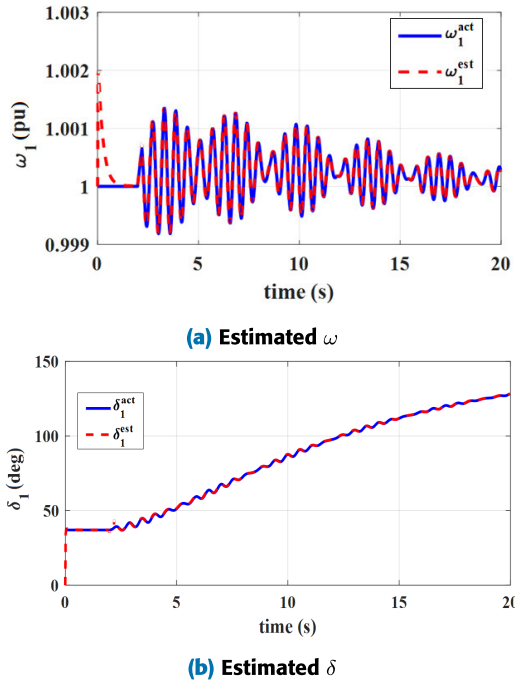


FIGURE 5. Estimated states using classical UKF DSE at bus 1.

loss of synchronization. The transient state is modelled by tripping the transmission line connected between bus-4 and bus-5 at $t = 2$ s and restoring it at $t = 2.2$ s. Fig. 6a shows the estimation of ω using the direct UKF DSE technique, whereas Fig. 6b shows the estimation of ω using the proposed DSE technique. The existence of a significant deviation can be observed between the estimated state from the direct UKF method and the actual state when the PMU data at bus 3 is missing. However, the proposed method is found to produce state estimates which are in line with the actual state giving an accurate state estimate even under the presence of missing data.

Fig. 7a shows the estimation of absolute rotor angle δ using the direct UKF DSE technique, whereas Fig. 7b shows the estimation of absolute rotor angle δ using the proposed DSE technique. It can be observed that the direct UKF algorithm shows a significant error from $t = 6$ s, when the PMU data is missing, whereas the proposed DSE technique estimates the state of the system efficiently in both cases. Hence, the robustness of the proposed technique is evident from the estimation.

C. CASE 3-TWO PMU MEASUREMENTS ARE MISSING

In this case, the data from the PMU's connected at bus-2 and bus-3 are purposefully cut off from time $t = 4$ s. In a real system, this case can originate due to the failure of two PMUs or the failure of the communication channel or PDC. Fig. 8 shows the simulation results for estimating ω at bus 2 using direct UKF based DSE technique and the proposed DSE algorithms.

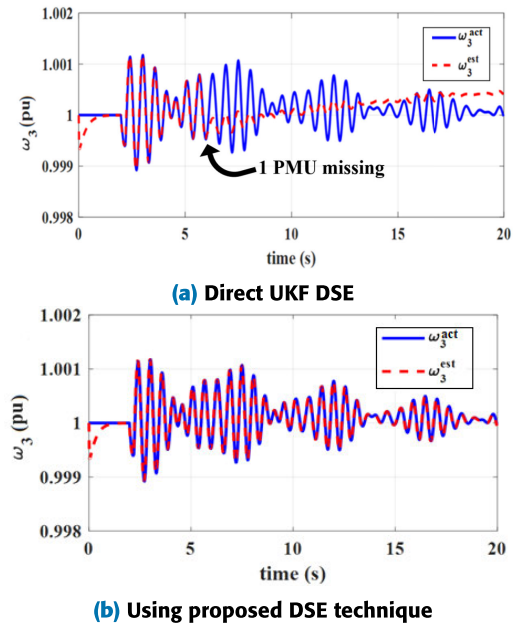


FIGURE 6. Estimated rotor speed ω at bus 3 using direct UKF DSE and proposed DSE technique.

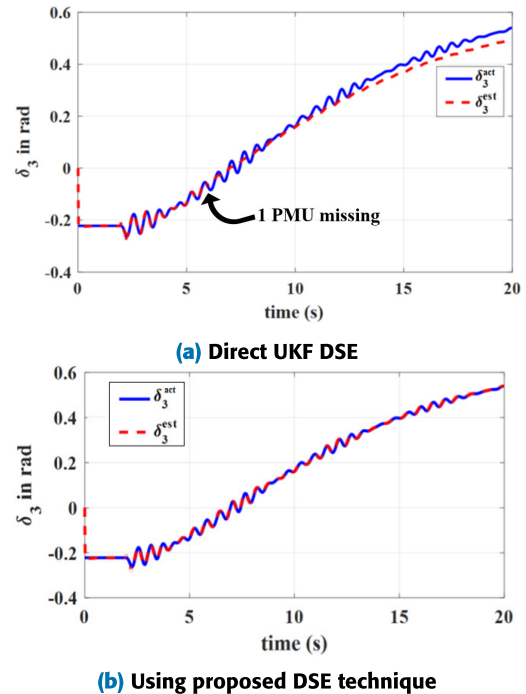


FIGURE 7. Estimated rotor angle δ at bus 3 using direct UKF DSE and proposed DSE technique.

Figure 9a and 9b shows the simulation results for estimating δ at bus 2 using direct UKF based DSE technique and the proposed DSE algorithms, respectively. It can be observed that the direct UKF algorithm shows a significant error from $t = 4$ s, when the PMU data is missing, whereas the proposed DSE technique estimates the state of the system efficiently in both cases.

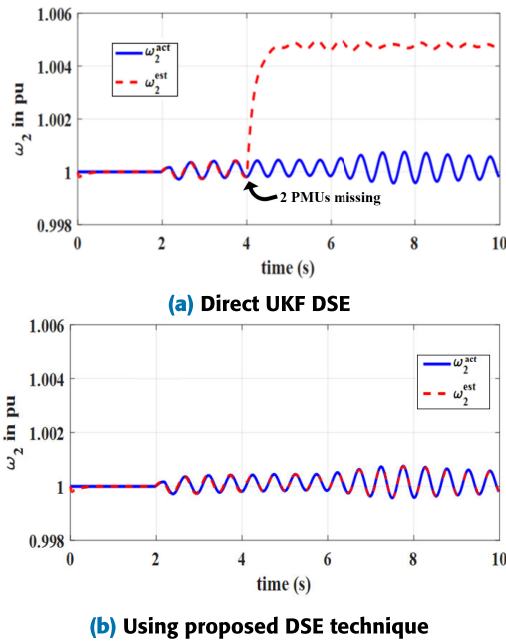


FIGURE 8. Estimated rotor speed ω at bus 2 using direct UKF DSE and proposed DSE technique.

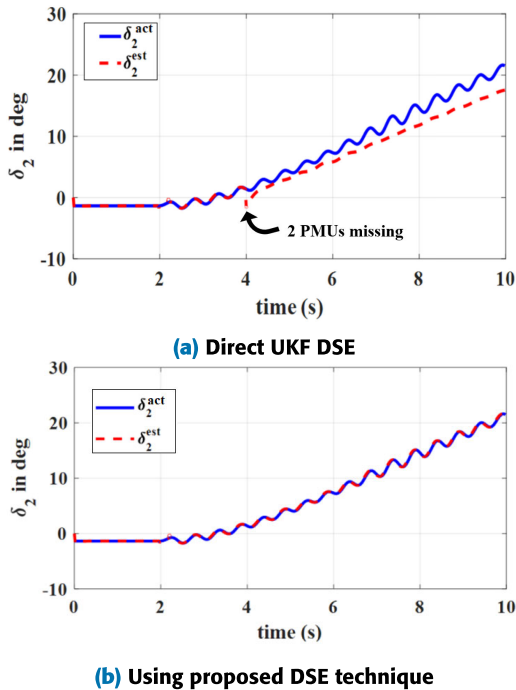


FIGURE 9. Estimated rotor angle δ at bus 2 using direct UKF DSE and proposed DSE technique.

V. PERFORMANCE OF PROPOSED METHOD

The proposed method estimates the state variables ω and δ under noisy as well as missing PMU conditions. To test the performance of the proposed method, the estimation errors for the obtained state estimates are calculated according

TABLE 2. Estimation errors.

Method	Normal Case		1 PMU Missing		2 PMU Missing	
	Error in ω (pu)	Error in δ (pu)	Error in ω (pu)	Error in δ (pu)	Error in ω (pu)	Error in δ (pu)
Direct UKF without De-noising.	0.000125	0.0029	0.0023	0.0200	0.0022	0.0447
Proposed Method without De-noising.	0.000125	0.0029	0.00013	0.0041	0.00017	0.0041
Direct UKF with De-noising.	0.000124	0.0028	0.0022	0.0189	0.00217	0.04434
Proposed Method with De-noising.	0.000124	0.0028	0.00012	0.0039	0.00016	0.00552

to (33)–(36). The estimation error [47] is given by

$$e_k = \frac{1}{2n} \sum_{i=1}^{2n} |x_k^{(i)} - \hat{x}_k^{(i)}| \quad (33)$$

where n represents the total number of state vectors, x_k^i represents the actual state vector at instant ‘ i ’ and $\hat{x}_k^i$ represents the predicted state vector using the UKF technique. The combined estimation error is given by where n represents the total number of generators connected in the system under consideration, x_k^i represents the actual state vector at instant ‘ i ’ and $\hat{x}_k^i$ represents the predicted state vector using the UKF technique. The combined estimation error is given by

$$e = \frac{1}{k_{\max}} \sum_{k=1}^{k_{\max}} e_k \quad (34)$$

The error in the estimated state variable ω is given by

$$e_\omega = \frac{1}{k_{\max}} \sum_{k=1}^{k_{\max}} \left[\frac{1}{n} \sum_{i=1}^n |\omega_k^{(i)} - \hat{\omega}_k^{(i)}| \right] \quad (35)$$

he error for the state variable δ is given by

$$e_\delta = \frac{1}{k_{\max}} \sum_{k=1}^{k_{\max}} \left[\frac{1}{n} \sum_{i=1}^n |\delta_k^{(i)} - \hat{\delta}_k^{(i)}| \right] \quad (36)$$

The estimation errors of both the techniques are compared and tabulated in Table 2

The estimation errors under normal case and the instance of missing PMU’s have been considered. The effect of de-noising has shown to reduce considerable error in the estimation. Also, it has been observed that the time taken for estimating the state is 0.00032s, even under the presence of the missing PMU Data or noise. The estimation errors highlight the efficiency of the proposed method.

VI. CONCLUSION

An effective DSE module that extracts the phasor data and estimates the state variables with a high level of accuracy even under the presence of missing PMU data or noisy measurements has been proposed in this work. The estimation is achieved without a substantial increase in the computation time. The PMU is modelled in MATLAB Simulink so that it can be used to obtain the real time PMU data that reflects the

dynamics of the power system. The data processing approach uses the WT based de-noising algorithm and the Modified UKF based DSE algorithm.

The PMUs are placed on an IEEE – 14 bus system for obtaining the PMU data which is then used for estimating the state variables of the synchronous generator. The simulation results are considered under the presence of noise in the PMU data and the complete absence of data from a PMU. Practically the PMU data can be missing due to the malfunctioning of the communication channel or PDC, or due to cyber attacks. The scenario of missing data from more than one PMU is also considered. The de-noising technique reduces the impact of error in the estimates, even though the UKF is able to rectify much of the error due to the noise. The simulation results highlight the effectiveness of the proposed model. Future work can focus on developing more advanced techniques to handle the extended periods of missing PMU data, potentially using predictive models or data reconstruction methods. The proposed method shall also be tested for the real time state estimation power systems integrated with converter based sources.

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India. His current area of research interest includes wide area monitoring of power systems.

SUBIN KOSHY received the B.Tech. degree in electrical and electronics engineering from the College of Engineering Trivandrum, Kerala, India, in 2011, and the M.Tech. degree in integrated power systems from the Visvesvaraya National Institute of Technology (VNIT), Nagpur, Maharashtra, India, in 2014. He has authored or co-authored two international book chapters and four international conference papers. He is also a co-inventor of a patent under the Government of



converters and AI based control algorithms for power electronics applications. He is a member of the IEEE Power Electronics Society.

V. SREEHARI VENUGOPAL (Member, IEEE) received the B.Tech. degree in electrical and electronics engineering from the NSS College of Engineering, Palakkad, Kerala, India, in 2013, and the M.Tech. degree in power systems from the National Institute of Technology Calicut, Kerala, in 2018. He is currently a Senior Research and Development Engineer with Delta Electronics India Pvt. Ltd., Bangalore, India. His current research interests include digital control of power



current research interests include the applications of wide area monitoring systems in power systems, power system stability and control, power system security, smart grids, and microgrids. She was a recipient of the POSOCO Power System Award (PPSA) for her research work, in 2014.

R. SUNITHA (Senior Member, IEEE) received the B.Tech. degree in electrical and electronics, the M.Tech. degree in energetics, and the Ph.D. degree in power systems from the National Institute of Technology Calicut, Kerala, India, in 1996, 1999, and 2014, respectively. She is currently an Associate Professor with the Department of Electrical Engineering, National Institute of Technology Calicut. She has authored and co-authored several research articles and book chapters. Her



technology Calicut, India. Her teaching and research interests include distributed generation and smart grid technologies.

ELIZABETH P. CHERIYAN (Senior Member, IEEE) was born in Alappuzha, India. She received the B.Tech. degree in electrical engineering from the University of Kerala, India, in 1996, and the M.Tech. degree in energetics from the National Institute of Technology Calicut, India, in 1998, and the Ph.D. degree from Indian Institute of Technology Bombay, India, in 2008. She recently retired as a Professor with the Department of Electrical Engineering, National Institute of Tech-



K. ANANTHAKRISHNAN (Student Member, IEEE) was born in Kozhikode, India. He received the B.Tech. degree in electrical engineering from the Government Engineering College, Barton Hill, Trivandrum, Kerala, India, under APJ Abdul Kalam Technological University, in 2023. He is currently pursuing the M.Tech. degree in power systems from the National Institute of Technology Calicut, Kerala. He is a Student Member of the IEEE Power and Energy Society.

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